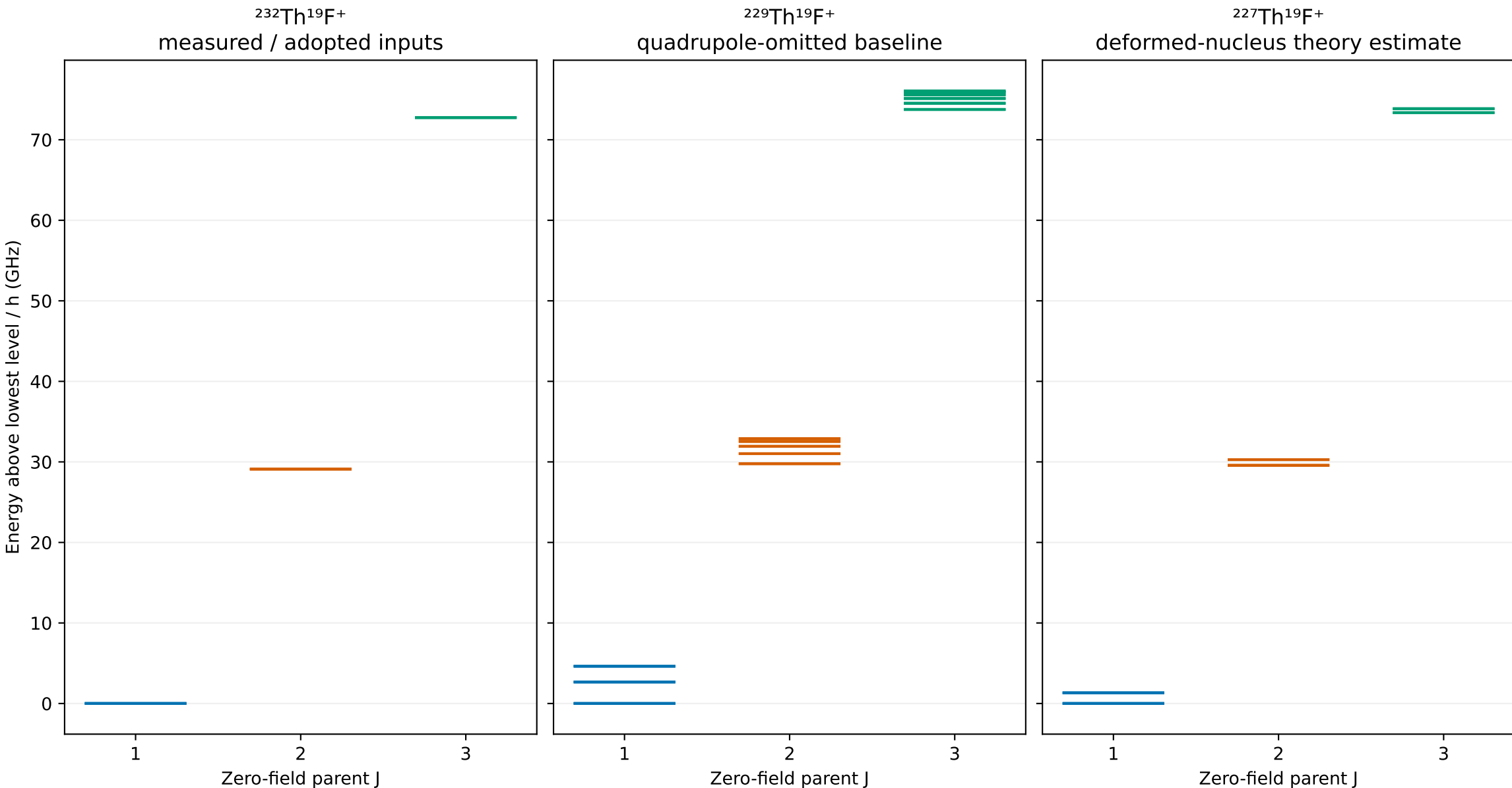


ThF⁺ X ³Δ₁ · zero-field rotational structure



Each isotope has its own zero. Odd-isotope molecular constants are unscaled transfers from ²³²ThF⁺.
 Unknown Th spin rotation omitted; ²²⁹Th quadrupole omitted. Exploratory models without physical uncertainty bands.